Lab 02 — ICMP Ping & Echo Analysis

Protocol: ICMP (Internet Control Message Protocol)
Tool: Wireshark on Kali Linux
RFC: RFC 792
Difficulty: Beginner
Time Taken: ~1 hour

Objective

Capture and analyze ICMP Echo Request/Reply packets generated by the ping command. Understand ICMP's role in network diagnostics, how it reveals network topology, and why it's heavily abused by attackers for reconnaissance, covert channels, and DoS attacks.

Protocol Background — Why ICMP Matters

ICMP operates at **Layer 3 (Network Layer)** and is used by network devices to send error messages and operational information. Unlike TCP/UDP, ICMP has **no port numbers** — it identifies message types by type codes.

ICMP is built into every IP stack. You cannot "turn off" ICMP entirely without breaking fundamental network diagnostics. This makes it a permanent feature that attackers have exploited since the early days of the internet.

ICMP Message Types (Key Ones)

Type	Code	Name	Use Case
0	0	Echo Reply	Ping response
3	0-15	Destination Unreachable	Host/port/network not reachable
5	0-3	Redirect	Route correction
8	0	Echo Request	Ping
11	0	Time Exceeded	TTL expired (used by traceroute)
13	0	Timestamp Request	Time sync

What I Captured

Basic Ping Capture

Command run on Kali:

```
ping -c 4 192.168.x.x   # Ping Ubuntu VM 4 times
```

This generates a pair of packets for each ping:

1. **ICMP Echo Request** (Type 8, Code 0) — sent by Kali

2. **ICMP Echo Reply** (Type 0, Code 0) — sent by Ubuntu

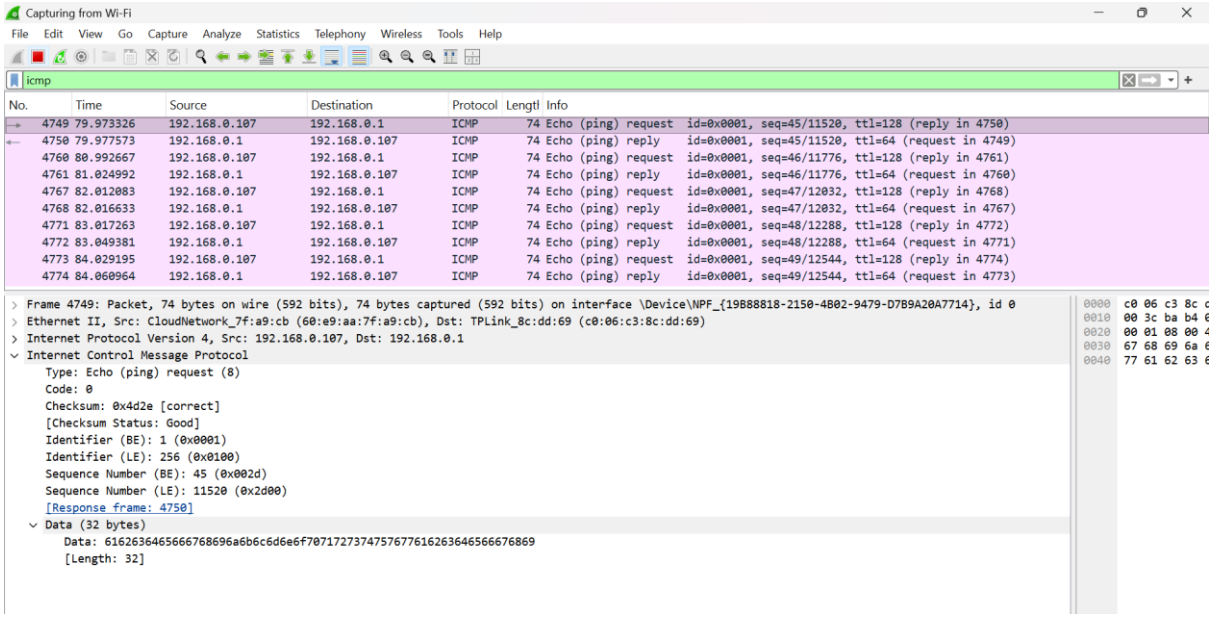
```
Kali (192.168.x.x)        Ubuntu (192.168.x.x)
      |                        |
      |—— Echo Request (Type 8) –► |
      | ◄—— Echo Reply   (Type 0) —|
      |                        |
      |—— Echo Request (Type 8) –► |
      | ◄—— Echo Reply   (Type 0) —|
```

Wireshark Filters Used

```
# All ICMP traffic
icmp

# Only Echo Requests (ping sent)
icmp.type == 8

# Only Echo Replies (ping received)
icmp.type == 0

# Destination Unreachable messages
icmp.type == 3

# Time Exceeded (traceroute behavior)
icmp.type == 11

# Filter ICMP from specific host
icmp && ip.src == 192.168.x.x
```

Screenshots & Findings

Screenshot 1 — ICMP Echo Request (Ping Sent)



What this shows:

- Kali sends an ICMP packet with **Type=8, Code=0** (Echo Request)
- Contains a **sequence number** — increments with each ping
- Contains an **identifier** — process ID of the ping command
- Payload: 56 bytes of data (pattern 0x10 0x11 0x12...) — this is the ping payload

Key fields in Wireshark:

Type: 8 (Echo Request)

Code: 0

Checksum: 0x[valid]

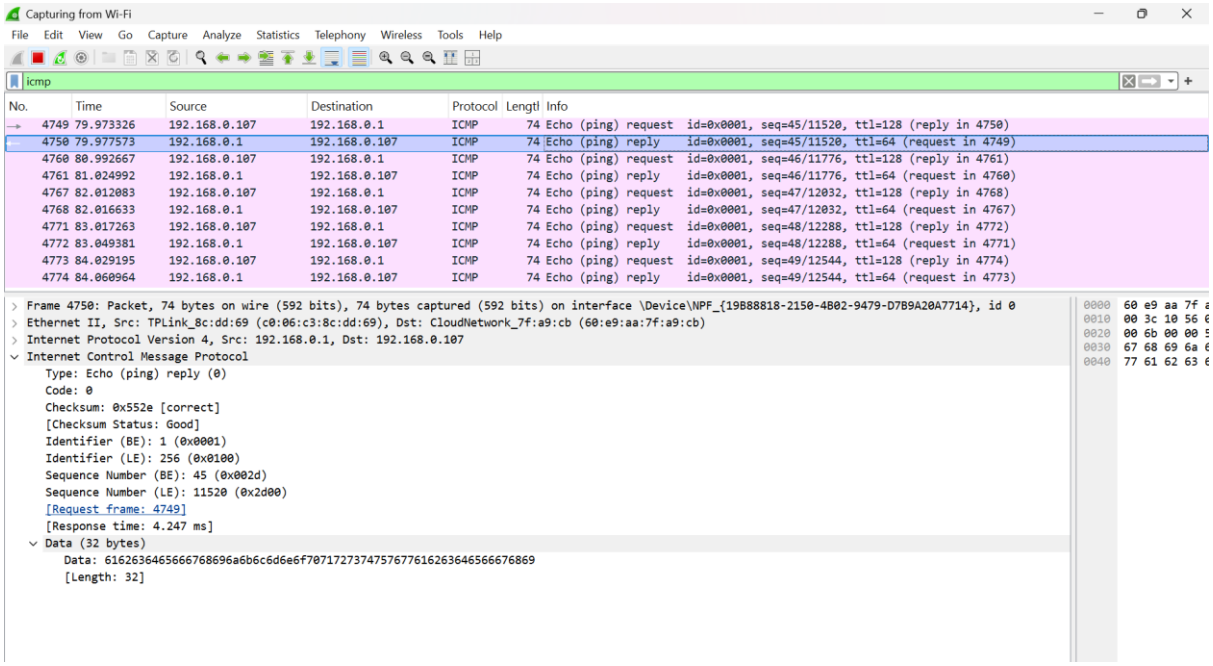
Identifier: [PID of ping process]

Sequence Number: 1 (increments each ping)

Data: 56 bytes

Important: The payload content in ICMP is controlled by the sender — attackers exploit this for **ICMP tunneling** (hiding data in the payload).

Screenshot 2 — ICMP Echo Reply (Ping Response)



What this shows:

- Ubuntu responds with **Type=0, Code=0** (Echo Reply)
- **Same sequence number and identifier** as the Request — this is how they're matched
- Same payload content echoed back
- **RTT (Round Trip Time)** calculated by comparing timestamps

Key fields:

Type: 0 (Echo Reply)

Code: 0

Identifier: [Same as request]

Sequence Number: [Same as request]

Why identifier + sequence matching matters: This prevents ICMP reply spoofing — a forged reply with wrong seq/identifier won't match and will be discarded.

What the TTL Field Reveals

One of the most powerful pieces of information in ICMP packets is the **TTL (Time To Live)** field in the IP header:

Starting TTL	Likely OS
64	Linux, macOS, Android
128	Windows
255	Cisco IOS, network appliances

Received TTL = Starting TTL − Number of hops

→ If received TTL = 63, the host is 1 hop away on Linux

→ If received TTL = 126, the host is 2 hops away on Windows

In Wireshark, look at: IP > Time to live field in the packet header.

This is **OS fingerprinting without sending a single TCP packet** — just ping responses reveal the target OS family.

Traceroute — ICMP Type 11 in Action

traceroute 8.8.8.8

Traceroute works by sending packets with **incrementing TTL values**:

- TTL=1 → dies at first router → router sends ICMP Type 11 (Time Exceeded) back
- TTL=2 → dies at second router → ICMP Type 11 from router 2
- ...continues until destination is reached

In Wireshark you'll see alternating:

→ UDP/ICMP with TTL=1

← ICMP Type 11 from Router 1

→ UDP/ICMP with TTL=2

← ICMP Type 11 from Router 2

→ UDP/ICMP with TTL=n (reaches destination)

← ICMP Type 0 (Echo Reply) from destination

This **maps the entire network path** between two hosts — extremely valuable for network recon.

Security Implications — What an Attacker Can Do

1. ICMP Ping Sweep (Network Reconnaissance)

An attacker sends ICMP Echo Requests to an **entire IP range** to discover live hosts — without touching any TCP ports:

```
# Nmap ping sweep

nmap -sn 192.168.1.0/24


# Direct ICMP sweep with fping

fping -a -g 192.168.1.0/24 2>/dev/null
```

In Wireshark: Hundreds of ICMP type-8 packets to sequential IPs in rapid succession — a clear sweep signature.

2. Ping of Death (Historical — CVE-1999-0128)

Sending an oversized ICMP packet (>65535 bytes) caused buffer overflows in old OS implementations. Patched in modern systems but illustrates how ICMP payload size can be weaponized.

3. ICMP Flood (Smurf Attack / DoS)

Attacker sends ICMP Echo Requests to a **broadcast address** with the victim's IP spoofed as the source:

Attacker → ICMP Request to 192.168.1.255 (broadcast), Src=Victim IP

All hosts on subnet → ICMP Reply to Victim IP

Victim receives hundreds of simultaneous replies — amplification DDoS.

Wireshark signature: Massive ICMP reply flood all destined for one IP.

4. ICMP Tunneling (Data Exfiltration) Advanced

Tools like **icmptunnel** and **ptunnel** hide TCP traffic (SSH, HTTP) inside ICMP payloads:

Normal ping payload: 0xAABBCCDD (padding)

Tunneled payload: [encoded TCP data / DNS query / command]

Detection: ICMP payload > 64 bytes is suspicious. Real pings rarely carry more than 56 bytes.

```
# Detect large ICMP payloads (potential tunnel)

icmp && data.len > 64
```

5. ICMP Redirect Attack

Attacker sends forged ICMP Type 5 (Redirect) to reroute victim's traffic through the attacker's machine — a man-in-the-middle without ARP spoofing.

Defender Perspective — Detection & Mitigation

Firewall Rules (Block External Ping Sweep)

```
# iptables — allow ICMP from internal network only
iptables -A INPUT -p icmp --icmp-type echo-request -s 192.168.0.0/16 -j ACCEPT
iptables -A INPUT -p icmp --icmp-type echo-request -j DROP

# Rate limit incoming ICMP
iptables -A INPUT -p icmp -m limit --limit 10/minute --limit-burst 20 -j ACCEPT
```

Detecting ICMP Tunnel in Wireshark

```
# Flag oversized ICMP
icmp.data_len > 100

# Unusual ICMP types
icmp.type != 0 && icmp.type != 8
```

SIEM Detection Logic

- Alert on > 50 ICMP requests/second from single source
- Alert on ICMP payload > 128 bytes
- Alert on ICMP to broadcast addresses

Key Takeaways

1. **ICMP reveals OS identity** via TTL values — even without TCP scanning
2. **Ping sweeps map live hosts** before any attack begins
3. **ICMP payload is attacker-controlled** — data can be hidden inside
4. **Traceroute exposes your full network topology** to anyone who can reach your network
5. **Blocking all ICMP breaks traceroute and diagnostics** — block selectively, not entirely

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References

- RFC 792 — Internet Control Message Protocol
- RFC 1122 — Requirements for Internet Hosts
- SANS Reading Room — ICMP Attacks Illustrated
- "The Art of Network Penetration Testing" — Royce Davis